



Interhospital conference

**A 7-year-old girl presented with fever with
Lt hemiparesis for 2 days**

Natchaya Kunanithaworn, MD.
23rd February 2024



Identification data

- เด็กผู้ป่วยหญิงชาวพม่าอายุ 7 ปี
- ภูมิลำเนา: ประเทศพม่า
- เข้ารับการรักษาที่รพ.มหาราชนครเชียงใหม่: วันที่ 13 ตุลาคม 2566
- แหล่งข้อมูล: ได้จากการซักประวัติผ่านล่าม



Chief complaint

- ไข้และอ่อนแรงซีกซ้าย 2 วันก่อนมารพ.



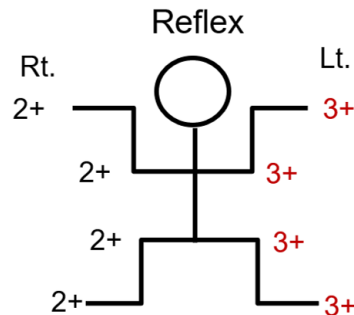
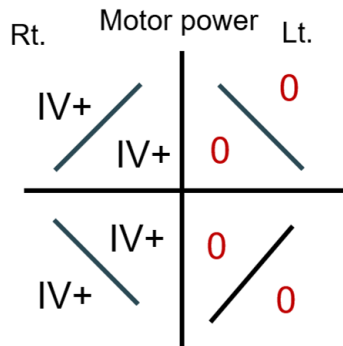
Present illness

2 วันก่อนมารพ. บิดาให้ประวัติว่าผู้ป่วยมีไข้ ไม่ได้วัดอุณหภูมิ บ่นปวดศีรษะ รู้สึกอ่อนเพลีย สังเกตว่านอนหลับตลอด ไม่สามารถลุกนั่งเองได้ ต้องพยุงลูก ไม่ยอมพูด รับประทานอาหารได้น้อย ขณะนั้นไม่มีอาเจียน ไม่มีถ่ายเหลว ไม่มีอาการไอหรือน้ำมูก ปฏิเสธอาการชักก่อนหน้านี้ บิดาพาไปตรวจที่รพ.อำเภอ



Physical examination แรกครั้งที่รพ.อำเภอ

- **Vital signs:** BT 39 °C, RR 20/min, BP 117/71 mmHg, HR 120/min, SpO2 98%
- **General appearance:** A girl, poor co-operation
- **HEENT:** no injected pharynx, no pale conjunctivae, no jaundice
- **Heart:** regular, no murmur
- **Lungs:** clear and equal breath sound both lungs
- **Abdomen:** soft, not tender
- **Neurological examination:**
 - E4V1M5, pupil 2 mm RTLBE, eye deviate to Rt., aphasia



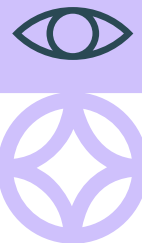
- **Stiffness of neck:** positive
- **Clonus** negative both feet
- **Babinski's reflex:** dorsiflexion both feet

Investigations แรกรับที่รพ.อำเภอ

CBC	Result
Hb (g/dL)	11
Hct (%)	35
WBC (cell/mm ³)	9,800
Neutrophil(%)	66
Lymphocyte(%)	22
Eosinophil(%)	1
Monocyte (%)	10
Platelet (cell/mm ³)	332,000

Blood chemistry	Result
BUN (mg/dL)	20
Cr (mg/dL)	0.48
Na (mmol/L)	139
K (mmol/L)	4.7
Cl (mmol/L)	107
HCO ₃ (mmol/L)	18
Ca (mg/dL)	9.2
Mg (meq/dL)	2.1
P (mg/dL)	4.4

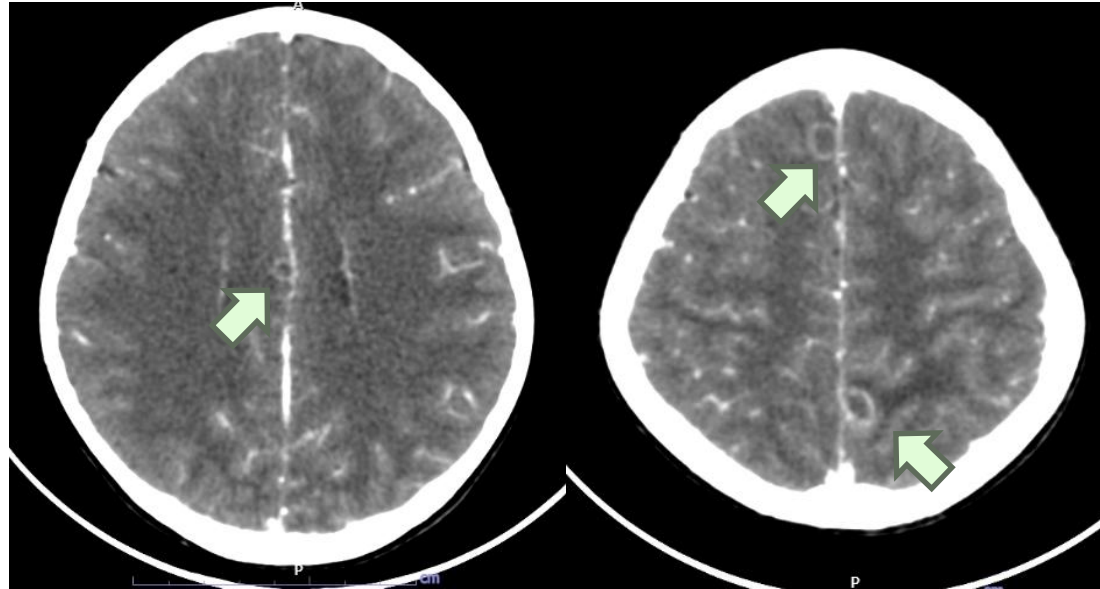
- ATK: negative
- Nasal swab for influenza: negative



CT brain with contrast ครั้งที่ 1 (Day 3, รพ.อำเภอก)

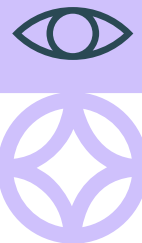
Findings:

- A few well-defined, rim-enhancing hypodense lesions with internal punctate enhancement and mild perilesional edema at body of corpus callosum, GM-WM junction of right frontal lobe and left parietal lobe
- DDx
 - Neurocysticercosis
 - Cerebral toxoplasmosis
 - Brain abscess or metastasis
- Suspicious diffuse leptomenigeal enhancement along bilateral cerebral sulci



CSF profile (รพ.อำเภอก)

CSF profile	Results
CSF protein	44 mg/dL (15-45)
CSF glucose	58 mg/dL (40-70)
CSF/plasma glucose ratio	0.57 (58/101) (POCT glucose 101)
Cell count and cell diff	WBC 63, low to diff, no RBC
CSF gram stain	no organism
CSF C/S	no growth in 2 days
Open pressure	not done



รพ.อำเภอ

- **Impression:** Acute meningitis with suspected multiple neurocysticercosis
- **Management**
 - Admit
 - Septic work up
 - On O2 cannula 2 LPM
 - Ceftriaxone (100 MKDay) 1g IV q 12 hr
 - Albendazole (15 MKDay) 150 mg oral bid pc
 - Dexamethasone (0.6 MKday) 3 mg IV q 6 hr
 - Dilantin (20 MKDose) 400 mg iv drip in 30 min then (5 MKDay) 50 mg IV q 12 hr



CSF profile: WBC 63, low to diff, no RBC, protein 44, glucose 58
CSF gram stain: no organism

CT brain with contrast ครั้งที่ 1:

- Multiple well-defined, rim-enhancing hypodense lesions with internal punctate enhancement and mild perilesional edema at body of corpus callosum, GM-WM junction of right frontal lobe and left parietal lobe



ผู้ป่วยเหนื่อยและซึมมากขึ้น → นึกถึงจาก Rt. lung atelectasis

V/S: RR 48/min, SpO2 81-85% (on O2 cannula)

- **GA:** comatose

- **Neuro:** E1V1M2, pupil 2 mm RTLBE

reflex - Lt upper and lower extremities → grade 0

- Rt upper and lower extremities grade 2+

Impression: Respiratory failure suspected from aspiration pneumonia with Rt lung atelectasis

- **Mx:** on ETT, CT brain with contrast emergency

CT brain with contrast ครั้งที่ 2:



Day3

Day of illness

Day5

Ceftriaxone (100 MKDay) IV q 12 hr

Albendazole (15 MKDay) 150 mg oral bid pc

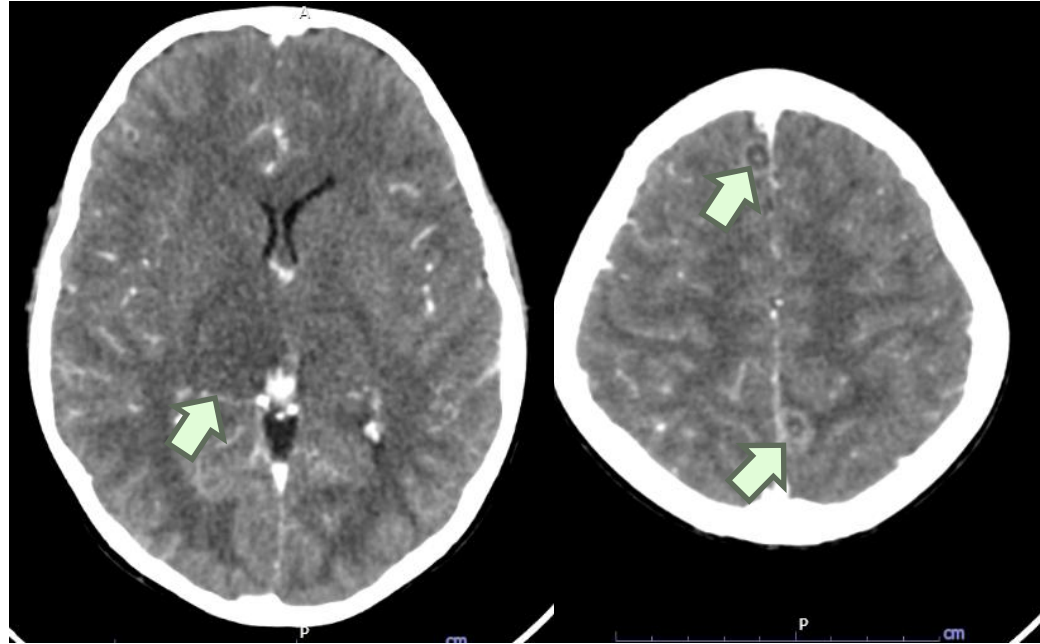
Management:

- Dexamethasone (0.6 MKday) 3 mg IV q 6 hr
- Dilantin (20 MKDose) 400 mg iv drip in 30 min then (5 MKDay) 50 mg IV q 12 hr

CT brain with contrast

ครั้งที่ 2 (รพ.อำเภอก)

- **More obvious diffuse hypodense area at the right thalamus**, edematous or infarction should be considered.
- **Overall decrease size but still seen perilesional brain edema of multiple ring enhancing lesions** with internal punctate calcification at corticomedullary junction of right frontal lobe and left parietal lobe, body of corpus callosum and right thalamus, the largest one recently measured 0.8x0.5 cm (previous 1.1x0.8cm).
- No leptomeningeal enhancement



Impression ที่รพ.อำเภอบ

- Acute meningoencephalitis with suspected multiple neurocysticercosis
- Hypodense lesion Rt. thalamus ddx acute ischemic stroke



CSF profile: WBC 63, low to diff, no RBC, protein 44, glucose 58
CSF gram stain: no organism

CT brain with contrast ครั้งที่ 1:

- Multiple well-defined, rim-enhancing hypodense lesions with internal punctate enhancement and mild perilesional edema at body of corpus callosum, GM-WM junction of right frontal lobe and left parietal lobe



ผู้ป่วยซึมและหายใจเหนื่อยมากขึ้น → นึกถึงจาก Rt. lung atelectasis
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- Rt upper and lower extremities grade 2+

Impression: Respiratory failure suspected from aspiration pneumonia with Rt lung atelectasis

- Mx: on ETT, CT brain with contrast emergency

CT brain with contrast ครั้งที่ 2:
hypodense area at the right thalamus



Refer CMU

Day8

Day3

Day5

Day of illness

Ceftriaxone (100 MKDay) IV q 12 hr

Albendazole (15 MKDay) 150 mg oral bid pc

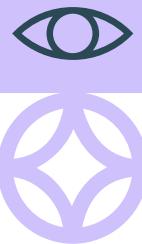
Management:

- Dexamethasone (0.6 MKday) 3 mg IV q 6 hr
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Past history

- คลอดที่บ้านที่ประเทศพม่า ไม่ได้รับวัคซีนตั้งแต่แรกเกิด
- ปฏิเสธโรคประจำตัว หรือยาที่รับประทานประจำ
- ก่อนหน้านี้แข็งแรงดี

อาการแรกเริ่มผู้ป่วยมีอาการตาเหลือกไปทางขวา มีแขนและขาอ่อนแรง
สองข้าง ไม่มีเกร็งกระตุก/ตาคะพริบ



Physical examination (CMU)

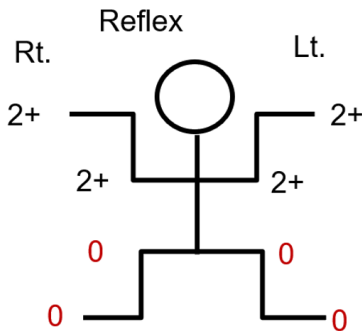
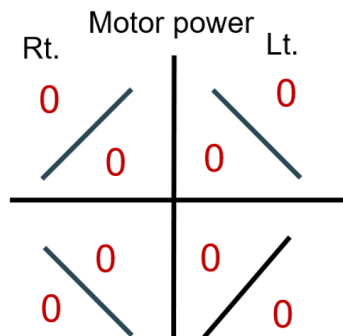
- **Vital signs:** BT 36.0 °C, RR 20 /min, HR 105 bpm, BP 120/50 mmHg, SpO2 98%
- **Physical examination:**
 - **GA:** on ETT with ventilator, awake but not aware
 - **HEENT:** not pale, anicteric sclerae
 - **Heart:** regular, normal S1&S2, no murmur
 - **Lungs:** clear and equal breath sound both lungs, no retraction
 - **Abdomen:** soft, no abdominal distension, liver and spleen not palpable, no guarding
 - **Skin:** no skin rash



Physical examination (CMU)

- **Neurological examination:**

- **E4VTM1**, pupil 3 mm slightly RTLBE, eye deviate to Rt.
- **Motor power grade 0 all extremities, hypotonia**
- Sensory: cannot evaluated
- Stiffness of neck: negative



- **Abdominal reflex: absent,**
- **Bulbocavernosus reflex: absent**
- **Loose sphincter tone**
- Clonus negative both sides
- Babinski's reflex: dorsiflexion both sides

CSF profile: WBC 63, low to diff, no RBC, protein 44, glucose 58
CSF gram stain: no organism

CT brain with contrast ครั้งที่ 1:

- Multiple well-defined, rim-enhancing hypodense lesions with internal punctate enhancement and mild perilesional edema at body of corpus callosum, GM-WM junction of right frontal lobe and left parietal lobe



ผู้ป่วยซึมและหายใจเหนื่อยมากขึ้น → นึกถึงจาก Rt. lung atelectasis

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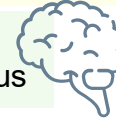
reflex - Lt upper and lower extremities → grade 0

- Rt upper and lower extremities grade 2+

- Mx: on ETT, CT brain with contrast emergency

CT brain with contrast ครั้งที่ 2:

hypodense area at the right thalamus



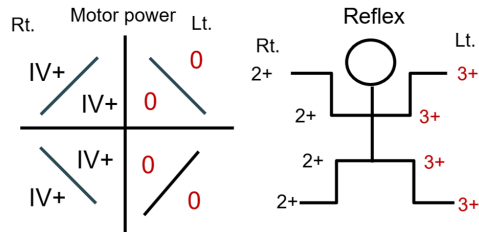
CMU

Day8

Day3

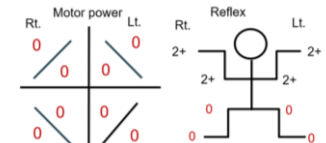
Day5

Day of illness



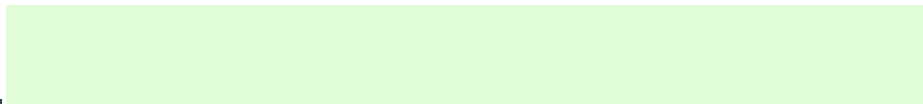
Neurological examination:

- E4VTM1, pupil 3 mm slightly RTLBE, eye deviate to Rt.
- Motor power grade 0 all extremities, hypotonia
- Sensory: cannot evaluated
- Stiffness of neck: negative
- Clonus negative both sides, Babinski's reflex: dorsiflexion both sides
- Abdominal reflex: negative, Bulbocavernosus reflex: negative





History taking and problem lists?



Problem lists



Problem lists

A 7-year-old girl presented with fever with headache for 2 days

- **Acute meningoencephalitis with cord involvement**
 - Lt hemiparesis with hyperreflexia with aphasia
 - meningeal sign irritation with alteration of consciousness
 - Thalamic involvement
 - Progress to quadriparesis with areflexia at both lower extremities
- **Suspected multiple neurocysticercosis with partial treatment**
- **Incomplete vaccination**



Differential diagnosis



Infectious cause

- **Virus**
 - Japanese encephalitis
 - Dengue virus
 - Influenza virus
 - Coronavirus
 - Enterovirus-71 (EV-71)
 - Poliovirus
 - EBV
- **Rickettsia**
 - Scrub typhus
- **Bacterial** (with complication)
 - Mycoplasma
 - *Streptococcus pneumoniae*
 - *Hemophilus influenzae*
 - *Neisseria meningitidis*
- **Mycobacterial**
 - *Mycobacterium tuberculosis*

Differential diagnosis

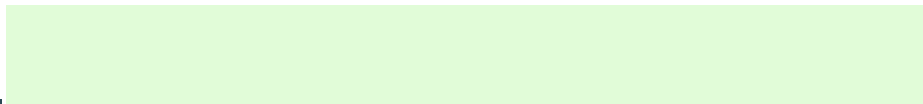
Non-infectious cause

- Autoimmune: NMO, MOGAD
- Vascular: venous sinus thrombosis

NMO; Neuromyelitis Optica Spectrum Disorders
MOGAD; Myelin Oligodendrocyte Glycoprotein-associated Disorders



Initial management?



Initial management

- Continue ceftriaxone (100 MKDay) IV q 12 hr
- Continue albendazole (15 MKDay) po bid
- Add azithromycin (10 MKDose) po OD
- Add acyclovir (30 MKDay) IV q 8 hr





Investigations?



Investigations



CT brain with contrast ครั้งที่ 3 (Day 10)

Findings:

- Ill-define hypodense lesions involving bilateral thalami and midbrain.
- The differential diagnosis included viral encephalitis, autoimmune encephalitis, hypoxic encephalopathy or artery of Percheron infarction.
- No abnormal enhancing lesion or leptomeningeal enhancement
- Well-opacified dural venous sinuses and cortical veins
- No intracranial hemorrhage

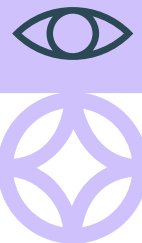


CSF profiles

CSF profile	รพ.อัมเภอ (day 3)	CMU (day 11)
CSF protein (mg/dL)	44	97 (15-45)
CSF glucose (mg/dL)	58	68 (40-70)
CSF/ Plasma glucose ratio	0.57 (58/101)	0.56 (68/120)
Cell count and cell diff	WBC 63 cells/mm ³ , low to differentiate	WBC 1 cells/mm ³ , RBC 0
CSF gram stain	No bacteria found	No bacteria found
CSF C/S	No growth in 2 days	No growth in 2 days

CSF for viral meningitis panel multiplex real time RT-PCR

Viral meningitis panel multiplex real time RT-PCR	Day 22 of illness
Human adenovirus	Not detected
HSV-1	Not detected
HSV-2	Not detected
VZV	Not detected
Epstein-Barr's virus (EBV)	Not detected
CMV	Not detected
Human herpesvirus 6 (HHV-6)	Not detected
Human herpesvirus 7 (HHV-7)	Not detected
Human parvovirus B19	Not detected
Human enterovirus	Not detected
Human paraechovirus	Not detected
CSF profile	Day 11 of illness
CSF PCR for herpes	Negative



Serum for JE virus

รายการทดสอบ	ผลการทดสอบ	วิธีทดสอบ
การตรวจหาแอนติบอดีต่อไวรัสเจีและเดงกีด้วยเทคนิค ELISA	Anti-Japanese encephalitis IgM 81 Unit (ค่าปกติ 0-39 unit)	Antibody capture ELISA
	Anti-dengue IgM 0 Unit (ค่าปกติ 0-39 unit)	
	Anti-dengue IgG 5 Unit (ค่าปกติ 0-99 unit)	

- Serum for JE IgM: **81 unit (0-39 unit)**
- ไม่ได้ส่ง CSF for JE virus

Viral serology

Virus	Day 8
Dengue IgM	Negative
Dengue IgG	Negative

Virus	Day 8
Herpes simplex IgM	Negative
Herpes simplex IgG	Negative
Varicella zoster IgG	Negative



Nasal swab

- Influenza type A (rapid test): negative
- Influenza type B (rapid test) : negative
- COVID-19 antigen test: negative



Stool for poliovirus (viral isolation)

- Negative

ลักษณะตัวอย่าง เหมาะสม		วิธีทดสอบ
รายการทดสอบ	ผลการทดสอบ	
การตรวจหาไวรัสโปลิโอ ด้วยเทคนิค Cell culture และ Real-time RT-PCR	แยกเชื้อแล้วได้ผลเป็นลบ	Viral Isolation
หมายเหตุ	1.ได้รับการรับรอง ISO 15189:2012 2.ตัวอย่างอุจจาระเก็บครั้งที่ 2	

IFA for scrub typhus

Virus	Results
Scrub typhus IgM	Negative
Scrub typhus IgG	Negative
Murine typhus IgM	Negative
Murine typhus IgG	Negative



Viral serology

Virus	Results
Mycoplasma IgM	Negative
Mycoplasma IgG	Negative



Anti-HIV test

- Non-reactive



Stool exam for parasite

- Negative



Investigations

	Results
ANA titer	Negative
Anti-dsDNA (IU/ml)	< 10



Sputum for AFB

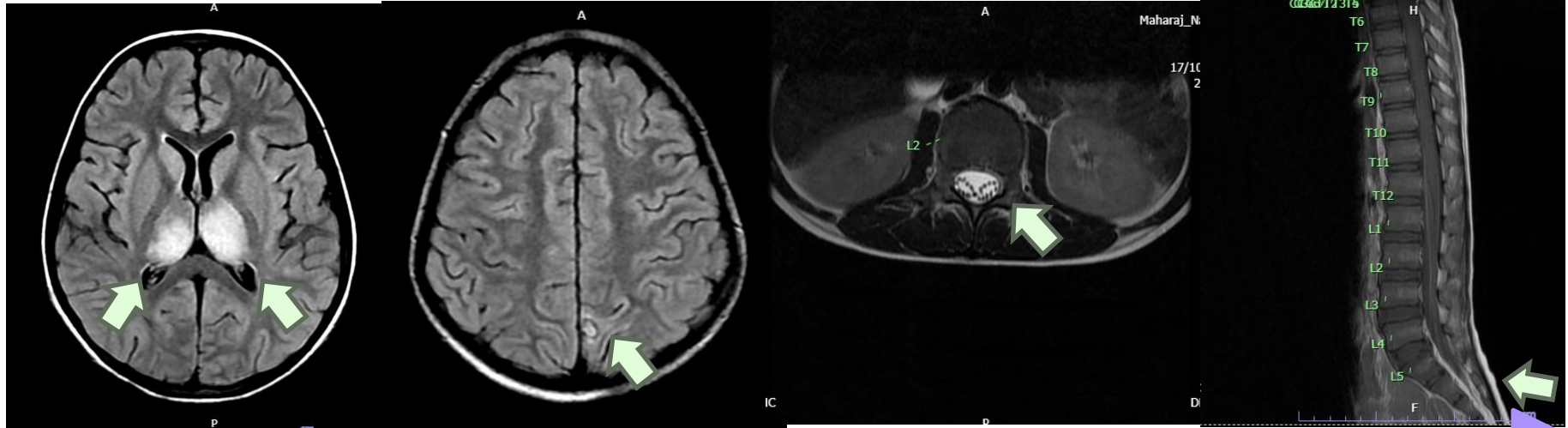
- Sputum AFB: negative* 3 days



MRI brain with whole spine (Day 12)

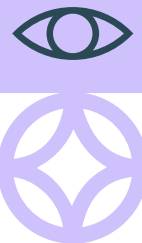
Impression:

- **Multiple T2-hyperintense signal lesions in bilateral thalami**, bilateral cerebral peduncles, right hippocampus and deep white matter of the left centrum semiovale/ left corona radiata with **enhancement of ventral cauda equina nerve roots; viral encephaloradiculitis** is first considered.
- Three rim enhancing lesions at parasagittal right frontal and left parietal lobes, corresponding with **granular nodular stage of neurocysticercosis**



CBC (แรกรับ CMU)

CBC	Day 5
Hb (g/dL)	10.7
Hct (%)	34.5
WBC (cell/mm ³)	11,990
Neutrophil(%)	81.6
Lymphocyte(%)	13.9
Eosinophil(%)	0.1
Monocyte (%)	4.3
Basophil (%)	0.1
Platelet (cell/mm ³)	364,000
MCV (fL)	68.6
RDW (%)	15.5



Management แกร็บที่รพ.มหาวิทยาลัยเชียงใหม่

- **RS system:**

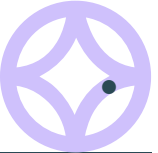
- On ETT with ventilator

- **CNS system:**

- Brain protective management
 - Keep PaO₂ > 80, PCO₂ 30-40, Na 140-150 meq/L, normal glucose, normal MAP
- Levetiracetam (40 MKDay) IV q 12 hr
- Dexamethasone (0.25 MKDose) IV q 6 hr
- Monitor EEG: no electrographic seizure

- **Medication:**

- Ceftriaxone (100 MKDay) IV q 12 hr → switch to meropenem (120 MKDay) IV q 8 hr
- Albendazole (15 MKDay) po bid
- Acyclovir (30 MKDay) IV q 8 hr
- Azithromycin (10 MKDose) po OD



CSF profile: WBC 63, low to diff, no RBC, protein 44, glucose 58
CSF gram stain: no organism
CSF C/S: no growth in 2 days

CT brain with contrast
ครั้งที่ 1

Day3

Day of illness

Respiratory failure suspected from aspiration pneumonia with Rt lung atelectasis
Hypodense lesion Rt. Thalamus

CT brain with contrast
ครั้งที่ 2

Day5

Refer CMU

Day8

CT brain with contrast
ครั้งที่ 3

Day 10



Ceftriaxone (100 MKDay) IV q 12 hr



Meropenem (120 MKDay) IV q 8 hr

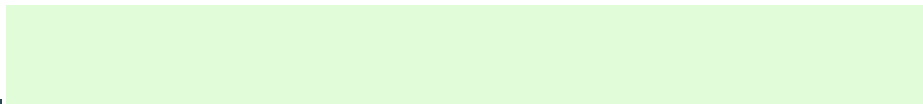
Albendazole (15 MKDay) 150 mg oral bid pc

Acyclovir (30 MKDay) IV q 8 hr

Azithromycin (10 MKDose) po OD

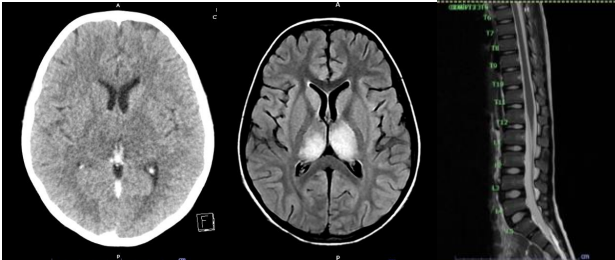


Progression

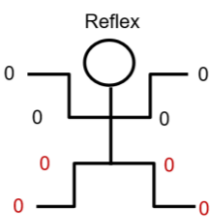


CSF profile: WBC 63, low to diff, no RBC, protein 44, glucose 58
CSF gram stain: no organism
CSF C/S: no growth in 2 days

Hypodense lesion Rt. Thalamus



PE: bicep reflex grade 0 both sides → areflexia
CSF: albuminocytologic dissociation
Imp: R/O GBS
Ix: NCV



CT brain with contrast ครั้งที่ 1



CT brain with contrast ครั้งที่ 2



CMU

CT brain with contrast ครั้งที่ 3



MRI brain with whole spine: radiculitis

Day3

Day5

Day8

Day 10

Day 12

Day 16

Day of illness

Ceftriaxone (100 MKDay) IV q 12 hr



Meropenem (120 MKDay) IV q 8 hr

Albendazole (15 MKDay) 150 mg oral bid pc

Acyclovir (30 MKDay) IV q 8 hr

Azithromycin (10 MKDose) po OD

Pulse Methylprednisolone 30 mg/kg/day IV

Progression

- **Nerve conduction study (Day 16):**
 - Finding of NCV consistent with Acute Motor Axonal Neuropathy (AMAN) (Subtype of Guillain-Barre syndrome)
- **CSF profile:** finding consistent with albuminocytologic dissociation



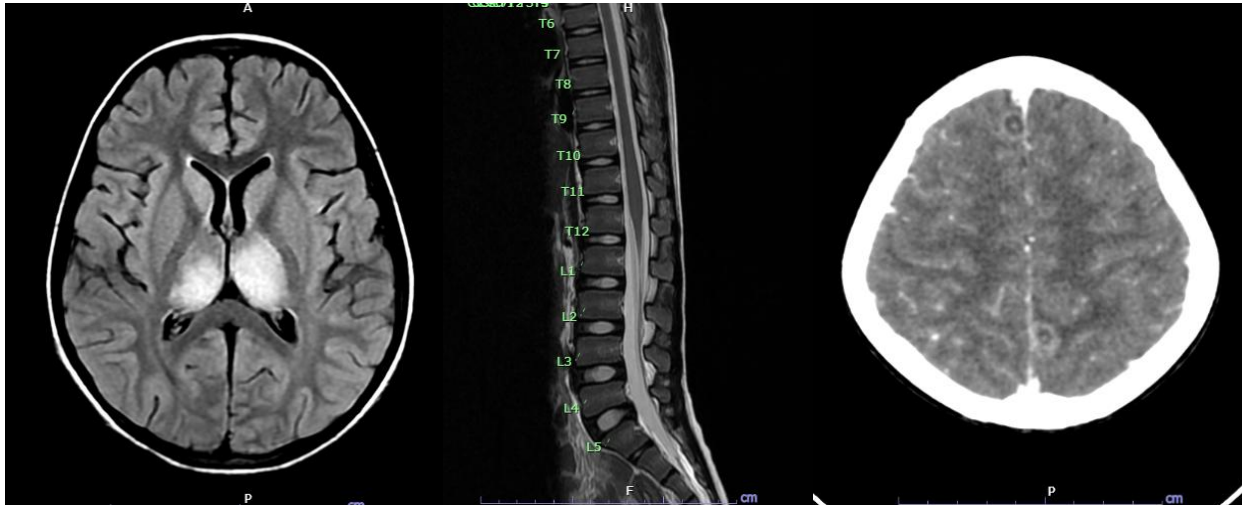
Comparison of TM and GBS

Characteristics	TM	GBS	Distinguishing Feature
Motor findings	Paraparesis or quadriparesis	Ascending weakness LE>UE in the early stages	TM: if UE involvement, often as severe as LE; often no UE involvement GBS: There usually is UE involvement and it is less severe than LE involvement early in the disease
Sensory findings	Usually can diagnosis a spinal cord level	Ascending sensory loss LE>UE in the early stages	TM: sensory level usually identified. Often no arm involvement GBS: no sensory level, usually UE less affected than LE early in the disease
Autonomic findings	Early loss of bowel and bladder control	Autonomic dysfunction of the cardiovascular (CV) system	TM: urinary urgency or retention early and prominent; CV instability only in severe cases higher than T6. GBS: urinary urgency or retention less common; CV instability is more common
Cranial nerve findings	None	EOM palsies or facial weakness	GBS: cranial neuropathies are more common than in TM
Electrophysiologic findings	EMG/NCV findings may be normal or may implicate the spinal cord: Prolonged central conduction on somatosensory evoked potential (SEP) latencies or missing SEP in conjunction with normal sensory nerve action potentials	EMG/NCV findings confined to the PNS: motor and/or sensory nerve conduction velocity reduced, distal latencies prolonged; conduction block; reduced H reflex usually present	The lack of peripheral nerve abnormalities in a patient with progressive weakness and sensory loss should suggest evaluation of the spinal cord for pathology Conversely, patients with suspected TM but equivocal clinical, lab or radiologic findings may warrant peripheral nerve examination
MRI findings	Usually a focal area of increased T2 signal with or without gadolinium enhancement	Normal	MRI abnormalities may be helpful in diagnosing a patient who is suspected of having GBS from TM
CSF	Usually, CSF pleocytosis and/or increased IgG index	Usually, elevated protein in the absence of CSF pleocytosis	CSF pleocytosis and elevated IgG index may be helpful in diagnosing a patient who is suspected of having GBS from TM

TM; Transverse myelitis
GBS; Guillain-Barré syndrome

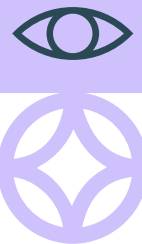
Final diagnosis

- Acute meningoencephalitis complicated with Guillain-Barre Syndrome (GBS) associated with JE virus infection
- Multiple neurocysticercosis (incidental findings)



Further management

- **Treatment for radiculitis**
 - IV methylprednisolone 30 mg/kg/day for total 5 days
 - **Guillain-Barre syndrome**
 - Consider role of IVIG → advice ญาติ เนื่องจากมีปัญหาระงับการจ่าย จึงไม่ได้ให้ IVIG
- Family meeting และแจ้งญาติเรื่อง prognosis
 - Advice เรื่องทำ tracheostomy → ญาติปฏิเสธ ไม่สามารถดูแลผู้ป่วยได้ ขอส่งกลับไป best supportive care รพ.ใกล้บ้าน



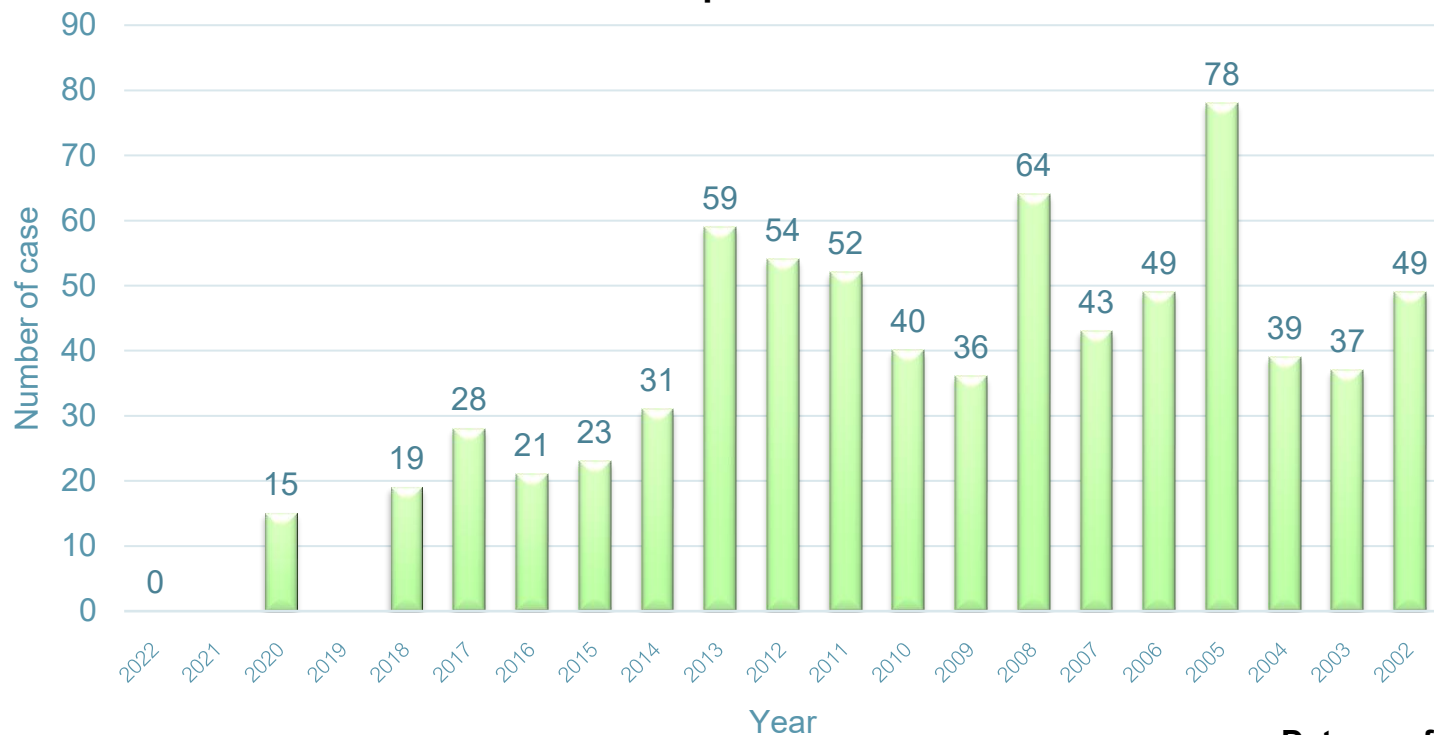


Japanese encephalitis



Update JE epidemiology in Thailand

Number of cases reported to the WHO



Data as of February 3, 2024

https://apps.who.int/gho/data/view.main.1540_42?lang=en

Japanese encephalitis

- Japanese encephalitis virus (JEV) is the most important cause of viral encephalitis in Asia.
- It is a mosquito-borne flavivirus, and belongs to the same genus as dengue, yellow fever and West Nile viruses.
- **Clinical manifestations:**
 - Most JEV infections are mild (fever and headache) or without apparent symptoms, but approximately 1 in 250 infections results in severe clinical illness.
 - **Severe disease:** rapid onset of high fever, headache, neck stiffness, disorientation, coma, seizures, spastic paralysis, meningoencephalomyelitis and ultimately death

Prognosis:

- Of those who survive, 20%–30% suffer permanent intellectual, behavioral or neurological sequelae such as paralysis, recurrent seizures

Japanese encephalitis

- **MRI brain:**

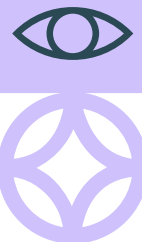
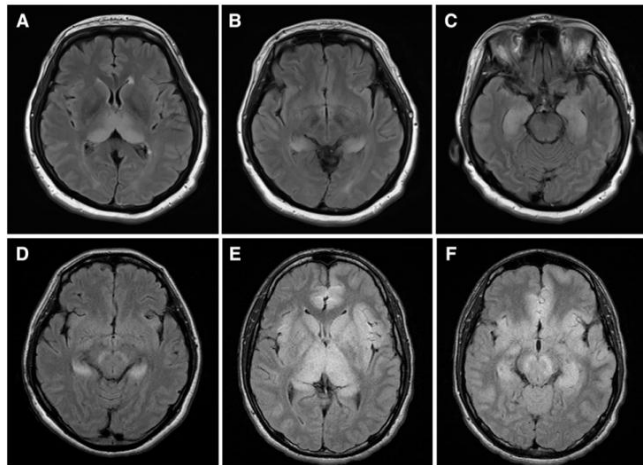
characteristically shows high-signal lesions in the thalamus, basal ganglia, midbrain, pons and medulla

- **Diagnosis:**

- **JEV-specific IgM antibody in cerebrospinal fluid (CSF) or serum**, using an IgM-capture ELISA.
- Testing of CSF sample is preferred to reduce false-positivity rates from previous infection or vaccination

Treatment:

- No antiviral treatment for JE
- Treatment is supportive to relieve symptoms and stabilize the patient.



Review literature

- A report from China recorded 47 adult patients with **Japanese encephalitis who presented with Guillain–Barré syndrome**
 - These patients had had symptoms compatible with JEV infection before the onset of the Guillain–Barré syndrome
 - All of 47 cases was detected JEV IgM in both serum and CSF
 - Albuminocytologic dissociation in the CSF was observed in 38/47 (81%) of these patients
 - Electromyographic results were consistent with the Guillain–Barré syndrome.
 - 22/47 cases of acute motor axonal neuropathy (AMAN) (47%)



Case summary

- A 7-year-old girl presented with fever with Lt hemiparesis for 2 days
 - Acute meningoencephalitis complicated with Guillain-Barre Syndrome (GBS) associated with JE virus infection
 - Multiple neurocysticercosis (incidental findings)
 - Incomplete vaccination



**Vaccination-
Preventable Diseases**



Thank you

